

Analysis Report

Group name: Group A

Group members: Shirley Wang, Xinyu Wang, Xueyun Bai, Stephanie Chen

Introduction

The goal of this project is to design a relational database system for the Last Resort Hotel. The system is intended to support major hotel operations, including room management, reservations, guest stays, events, billing, and access tracking. In addition to database design, we implemented an interactive web interface that allows hotel staff to perform operations such as searching room availability, managing check-in and check-out, viewing event schedules, and reviewing billing information. Through this system, hotel staff can manage complex workflows such as assigning rooms dynamically based on availability, organizing events, and handling billing for both individuals and organizations. In addition, the system records guest access through PIN-based cards, allowing the hotel to track room usage and improve security. The ERD developed in this milestone captures the main entities in the system and the relationships between them, providing a structured representation of how the hotel operates.

System Understanding

The Last Resort Hotel operates across multiple buildings, each containing several wings, floors, and rooms. Rooms serve different purposes, including sleeping rooms, meeting rooms, and suites. Guests make reservations by specifying preferences such as number of guests, bed types, or smoking preferences, but the actual room assignment may happen later depending on availability. Room availability can also change due to cleaning, maintenance, or renovation, so the system needs to track room status carefully.

In addition to regular stays, the hotel hosts events that require meeting rooms and are managed by designated hosts. These events may use rooms at different times of the day, such as morning or evening sessions. The system also supports flexible billing, where charges may be assigned to either individuals or organizations. Charges can be generated from stays, events, or additional services. Finally, the hotel uses access cards for room entry, and each usage is logged to track entry and exit times for both operational and security purposes. The system provides real-time access to this information through an interactive interface, allowing staff to quickly retrieve and analyze data when needed.

Key Design Decisions

First, the hotel location structure was modeled hierarchically using Building → Wing → Floor → Room, which reflects the physical organization of the hotel complex. This structure is represented through a set of one-to-many relationships connecting these entities.

Second, many-to-many relationships were resolved using associative tables. For example, the relationship between rooms and bed types is implemented through the room_bed table, which records the quantity of each bed type in a room. Similarly, event_room is used to represent rooms used in events, and room_reservation captures planned room allocations.

Third, reservations, planned room allocations, and actual room usage were clearly separated. A reservation represents the guest's intent to book rooms, room_reservation represents the planned allocation of rooms for a specific time period, and room_assignment represents the actual occupancy during a stay. This separation allows the system to distinguish between booking plans and real usage, and supports flexibility in room assignment.

Fourth, the system distinguishes between guests, hosts, and organizations using the Person entity and role-specific entities such as Guest and Host. The Host entity is directly linked to events to represent event organizers, while the Billing_Party entity is used to unify both individuals and organizations responsible for payment across reservations and charges.

Fifth, charges were designed to have a clear origin. Each charge is associated with either a stay or an event, ensuring that all charges can be traced back to a specific activity in the system.

Finally, the system was designed to support interactive access to data through a web application. Key operations such as room availability search, event lookup, billing review, and access tracking are implemented as interactive features rather than static queries. This design ensures that hotel staff can efficiently retrieve up-to-date information and respond to operational needs in real time.

Assumptions

Because the business description did not specify all details, several assumptions were made during database design. Room numbers are assumed to be unique within a wing, and wing names are assumed to be unique within a building.

Reservations represent requests for rooms, while actual room assignments may occur later depending on availability. In our design, reservation stores the booking request, room_reservation stores the planned room allocation, and stay with room_assignment stores the actual occupancy. This separates planned usage from actual room usage.

A room may contain multiple bed types, and the quantity of each bed type is stored in the `room_bed` table. Rooms may also have type-specific details depending on their function. Some rooms are modeled as sleeping rooms, some as meeting rooms, and suites are represented as special configurations that may reference both sleeping and meeting room components. Event rooms may be used during different time slots, such as breakfast, morning, lunch, afternoon, or evening. These time slots are represented by the attribute `usageSlot` in the `event_room` table.

Billing parties may be individuals or organizations, and each reservation must be associated with a billing party responsible for payment. Each charge is associated with either a stay or an event, but not both, so that every charge has a clear source.

Access cards are assigned to guests, and each card usage creates a log entry recording the room accessed and the time of access.

Tradeoffs and Limitations

The current design focuses on implementing the core functionality of the hotel system and ensuring that all major components, such as reservations, room assignments, events, billing, and access tracking, are correctly represented and connected. The system also includes an interactive web interface that allows users to perform key operations, such as searching availability, managing check-ins, and reviewing billing information.

One limitation is that advanced business logic, such as dynamic pricing, discounts, or tax calculations, is not implemented in the current system. These features would require additional application-level logic beyond the scope of this project.

Another limitation is that while the system includes additional visual and conceptual features on the interface, such as representations of dining, spa, and other resort services, these are included primarily for completeness of the hotel experience and are not fully implemented as functional modules, since they were not required by the project specification. These areas can be expanded in future work to support a broader range of hotel services.

In addition, although the system provides analytical reports, most outputs are presented in tabular form. More advanced data visualization, such as interactive charts and dashboards, could be added to improve usability and support decision-making.

Overall, the current system successfully demonstrates the core database design and interaction requirements, while leaving room for future extensions in business logic, service modules, and visualization.

Handling Ambiguities in the Description

The project description contained several ambiguous areas that required interpretation. The relationship between reservations and room assignments was not explicitly defined, so we separated the process into different stages. A reservation represents the guest's booking intent, room_reservation represents the planned room allocation, and room_assignment represents the actual room occupied during a stay.

Another ambiguity involved room types. Since rooms may serve different purposes, we used separate detail tables for sleeping rooms, meeting rooms, and suites. Suites are treated as special room configurations that may reference both sleeping and meeting room components. Billing responsibilities were also unclear in the description. To handle this, we created the billing_party entity to represent either an individual or an organization. We also defined each charge as belonging to either a stay or an event, which makes the source of every charge clear. The description of event room usage did not specify how time periods should be recorded, so we modeled this using the usageSlot attribute in the event_room table to represent predefined time periods.

Team Planning

Our team worked collaboratively throughout the project. We began by discussing and analyzing the system description together to ensure a shared understanding of the requirements. All team members contributed to the ERD design, database implementation, SQL queries, and web application development. We also worked together to refine the interface design, test the system, and prepare the final report and presentation. This collaborative approach helped ensure consistency across all components of the project.